

The Fidelity-Utility Paradox in Surrogate-Based RL for HVAC Control

A more accurate surrogate can be a worse RL training environment — it exposes a rougher action→temperature surface that the policy exploits into failure on the real building. A role-separating hybrid keeps the surface smooth and the controller robust.

Surrogate training environment

Coarse black-box v3
(1 h step — less accurate)

Accurate twin
(calibrated v3.5 / fine-res v3)

Hybrid
(v3 rollout + frozen
v3.5 censor)

Action → next-temperature response surface (measured)

smooth · monotone

action a_0

rough · non-monotone

action a_0

*smooth (v3 dynamics)
+ plausibility censor*

action a_0

Controller on the live building

✓ **USABLE**

<5% comfort violation

✗ **COLLAPSE**

>77% violation, $m_s > 1$

✓ **ROBUST**

<5% violation, ~85× faster